MVLU COLLEGE

Subject:-Artificial Intelligence

Aim:-Feed Forwading

Code:-

T098.py - C:/Users/vikas/OneDrive/Desktop/AI/T098.py (3.14.6)

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```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
df = pd.read_csv("Iris.csv")

# Select input features and target variable
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Standardize the input features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create the feed forward neural network
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
```

Activate Windows
Go to Settings to activate Windows.

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```
)

# Train the neural network using backpropagation
model.fit(X_train, y_train)

# Predict the classes for test data
y_pred = model.predict(X_test)

# Calculate test accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Test Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Display the confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Visualize the confusion matrix
plt.figure(figsize=(6, 5))

plt.imshow(cm)
plt.title("Neural Network Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.colorbar()

classes = model.classes_

plt.xticks(range(len(classes)), classes, rotation=20)
```

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T074 Krishnn

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```
print(clSSassification_report(y_test, y_pred))

# Visualize the confusion matrix
plt.figure(figsize=(6, 5))

plt.imshow(cm)
plt.title("Neural Network Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.colorbar()

classes = model.classes_

plt.xticks(range(len(classes)), classes, rotation=20)
plt.yticks(range(len(classes)), classes)

# Display values inside the matrix
for i in range(len(classes)):
    for j in range(len(classes)):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.tight_layout()
plt.show()

# Display training loss curve
plt.figure(figsize=(8, 5))

plt.plot(model.loss_curve_)

plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.title("Backpropagation Training Loss")
plt.grid(True)

plt.show()
```

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Output:-

IDLE Shell 3.14.6

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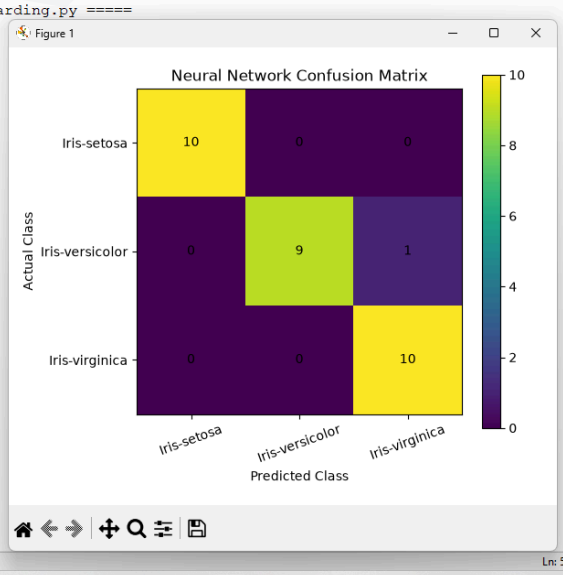
```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
====== RESTART: C:/Users/vikas/OneDrive/Desktop/AI/T098 Feed Forwarding.py =====
Test Accuracy: 0.9666666666666667
Accuracy Percentage: 96.66666666666667 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]

Classification Report:
                precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        10
Iris-versicolor       1.00      0.90      0.95        10
 Iris-virginica       0.91      1.00      0.95        10

       accuracy                           0.97        30
      macro avg       0.97      0.97      0.97        30
   weighted avg       0.97      0.97      0.97        30
```



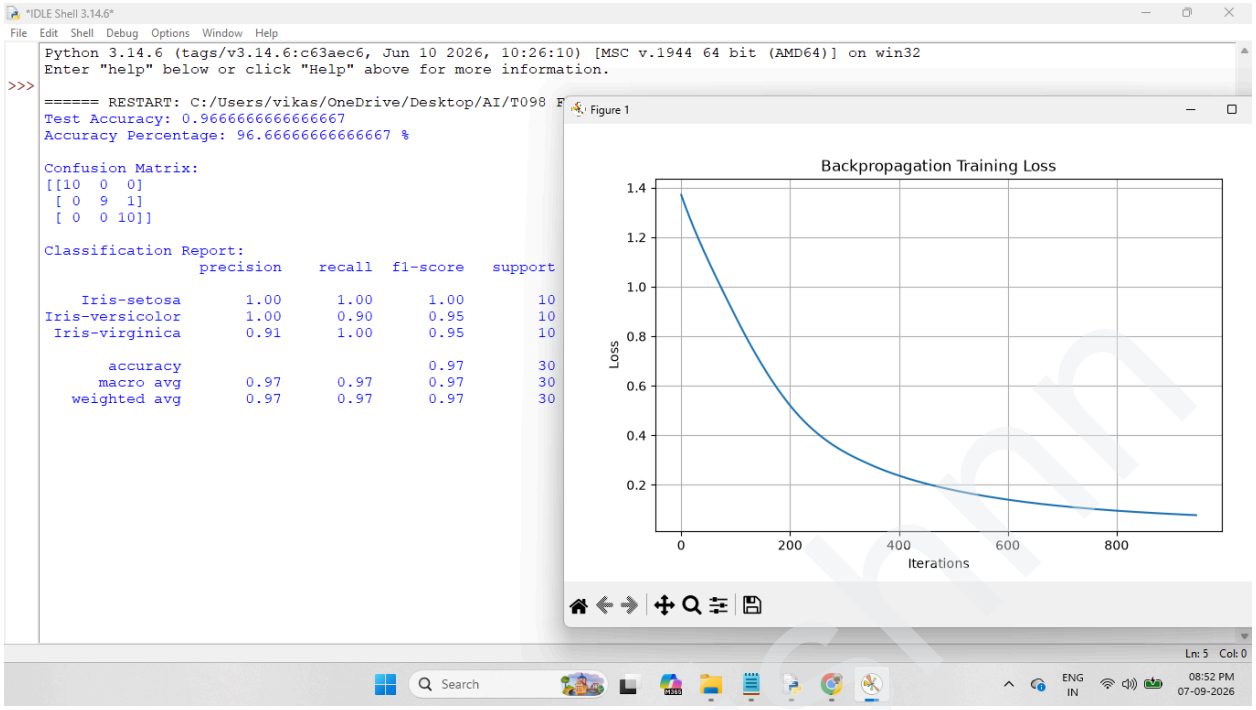
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